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RESEARCH ARTICLE

Acute Effects of High-Intensity Interval Training (HIIT) on Heart Rate Variability and Psychoaffective Responses Among Bipolar Patients and their Control Peers: A Randomized Crossover Trial

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Abstract: ***Introduction***: to evaluate the acute effects of continuous moderate-intensity aerobic exercise and high-intensity interval training (HIIT) on heart rate variability (HRV) and psychoaffective responses in patients with bipolar disorder (BD) and healthy controls.

Methods: Eight BD patients and eight controls underwent baseline assessments, including anthropometric measurements and a submaximal exercise test to determine VO_2Max, followed by two randomized exercise sessions. In one session, participants performed 12 min of continuous exercise at 65%VO_2Max, and in the other, during the HIIT protocol, participants engaged in six 45-s bouts at 100%VO_2Max, each followed by a 1-min and 15-s recovery period at 40%VO_2Max. Pre- and post-exercise measurements included psychoaffective scales (feeling scale - FS, felt arousal scale - FAS, and SUDs anxiety scale) and HRV assessments recorded during a 10-minute rest period under controlled conditions.

Results: Regarding RMSSD, both participants and sessions demonstrated a significant increase in this indicator (p=0.012). The BD significantly increased VLF and LF values and reduced HF values for both exercise sessions. Meanwhile, the LF/HF ratio showed a significant increase only in the HIIT session. For the control group, only a significant reduction in the HF index in both sessions and an increase in the LF/HF ratio only in the HIIT session. The FAS showed significant increases in bodily activation post-exercise across groups and modalities, while the FS demonstrated significant positive shifts in affective valence, with larger improvements following HIIT. Both exercise protocols produced significant reductions in anxiety levels (p=0.001), with HIIT showing a trend toward superior anxiolytic effects in BD patients.

Discussion: The improvement in vagal tone and HRV modulates the function of critical brain regions involved in emotional regulation, such as the prefrontal cortex, amygdala, and anterior cingulate cortex. This modulation reduces limbic hyperactivity and impulsivity while influencing monoaminergic pathways, attenuating excessive dopaminergic signaling, thereby contributing to enhanced mood stability and the reduction of affective disturbances in individuals with BD.

Conclusion: Both continuous moderate-intensity exercise and HIIT elicited favorable acute autonomic and psychoaffective responses in BD patients and healthy controls. Notably, HIIT tended to produce greater affective and anxiolytic benefits in individuals with BD.

Keywords: Autonomic function, affective responses, aerobic exercise, high intensity interval training.

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1. INTRODUCTION

Bipolar disorder (BD) is a serious mental disorder associated with intense mood fluctuations (mania and euthymia), which has a relatively low prevalence (1-3%); however, this population has a lifetime mortality rate due to suicide of approximately 15-20% [1-3], which, because it is a death, can be stratified as high and, therefore, worrying.

An interesting systematic review study with meta-analysis demonstrated that individuals with BD appear to have reduced heart rate variability (HRV) compared to their control peers [4]. Furthermore, studies suggest that this reduction in HRV becomes more pronounced as the disease worsens [5, 6]. It is worth noting that, in mania, sympathetic autonomic indicators are more active compared to euthymic moments. Thus, it is suggested that HRV may be an important prognostic indicator for patients with BD [7].

Studies suggest that regular aerobic exercise may increase HRV in patients with BD, indicating an improvement in sympathovagal balance, which may have an impact on mood stability, which in turn may result in fewer episodes of mania and thus improve the quality of life of these patients [8-10]. Although the measurement of HRV has already been observed in an incipient way within the BD spectrum at different moments of the cycle (depression, mania, euthymia) [8, 11], the acute effects of aerobic exercise on HRV components in these patients still need to be further investigated. Furthermore, the application of different aerobic intervention models (exercise program configuration), such as interval exercise, on HRV indicators and mood state, to the best of our knowledge, has not yet been investigated. It is noteworthy that a preliminary, albeit relatively recent, investigation indicates that high-intensity interval training (HIIT) constitutes a safe and well-tolerated therapeutic approach for individuals with mental disorders, with potential benefits extending to the attenuation of stress, anxiety, and depressive symptoms, as well as improvements in the Psychological General Well-Being Index [12].

Given the above, the objectives of the present study are: 1) to investigate the acute effects of high-intensity interval aerobic exercise (HIIT) on HRV indices (time and frequency domain), as well as on affective responses (feeling scale); 2) to track, using the circumplex model, the affective responses to HIIT and; 3) to analyze the anxiety responses to different types of exercises among patients with BD and their control peers. Finally, 4) to investigate the level of association between performance variables and affective responses.

As the main hypothesis (H1), we believe that HRV-related variables will show differences before and after the intervention and between patients with BD and their control peers. In addition, positive affective responses for both participants and exercise models will be observed, however, with a greater magnitude for HIIT. As a secondary hypothesis (H2), affective responses will transit in positive quadrants of affect, equally impacting the anxiety levels of the participants. Finally, there will be a significant association between the affect variables and aerobic performance.

2. METHODS

2.1. Study Design

The present study used as a reference the assumptions described by the "International Committee of Medical Journal Editors" (ICMJE) and respected all the items proposed in the guideline "*CONSORT 2010 statement: extension to randomised crossover trials*" (CONSORT Statement) for reporting randomized crossover trials [13]. The study was conducted throughout 2017, with a recruitment period of two months and a total duration of five months, and the total exposure of patients did not exceed three weeks of intervention. All patients were part of a follow-up group at a federal outpatient clinic located in the South Zone of Rio de Janeiro. Resolution 466/2012 of the National Health Council was followed, as well as approval by the institutional research ethics committee no. (Universidade Federal do Rio de Janeiro - #045.2015). All procedures were performed in accordance with the Declaration of Helsinki.

Three visits were made to the laboratory. All recruited patients had a previous diagnosis of BD and were being monitored at the federal outpatient clinic of the University. All patients underwent prior medical evaluation in a location adjacent to the research laboratory, where BD symptom scales were applied for subsequent classification according to the study characteristics. The control participants, who were not bipolar, did not undergo this stage and, therefore, began data collection immediately. After inclusion designation by the medical team, patients with BD and their non-bipolar control peers underwent a battery of anthropometric procedures, blood pressure measurement, as well as a submaximal exercise test to obtain VO_2Max and workloads at the first visit. At subsequent visits (two and three), subjects were randomly assigned to perform a continuous or HIIT exercise session. During the experimental sessions, baseline measurements were performed in the following order: a) feeling scale; b) felt arousal scale; c) SUDs anxiety scale. After the initial measurements, all participants rested for ten minutes and then had their HRV measurements obtained. To do so, all individuals were required to remain blindfolded for ten minutes (sitting in a reclining chair in a comfortable position and in a dark environment) and breathing spontaneously. Immediately afterward, the exercise procedures began (protocol described in the exercise session). Exactly five minutes after the end of the exercise procedures, a new HRV measurement was performed under the same conditions as the pre-exercise measurement, also for a period of 10 minutes. Five minutes after analysis, all scales (a) feeling scale; b) felt arousal scale; c) SUDs anxiety scale) were answered again in the same order. Fig. (**1**) represents the experimental research design.

2.2. Sample

The sample consisted of 16 participants: eight control individuals, *i.e.*, apparently healthy individuals, and eight patients diagnosed with BD (in the euthymic phase). Both the control and patients with BD were stratified as controls [14]. Participants were invited through calls made at the Specialized Outpatient Clinics of the Institute of Psychiatry

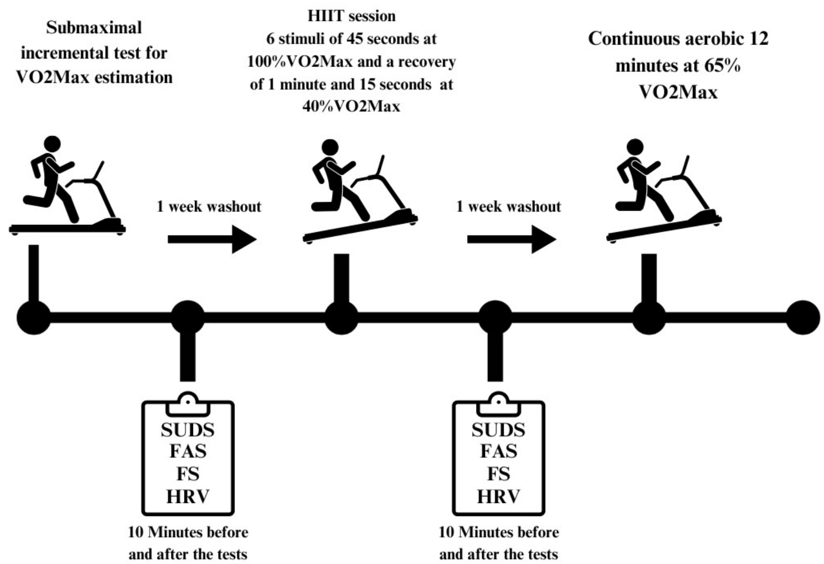

Fig. (1). Experimental design. (*A higher resolution/colour version of this figure is available in the electronic copy of the article*).

of UFRJ (IPUB/UFRJ). As an inclusion criterion, patients diagnosed with BD type II were included in the study, according to the criteria of the "Structured Clinical Interview for DSM-IV [15] and American Psychiatric Association, with four women and four men in each group, aged between 30 and 50 years, and who were controlled by medication (Described in item 3.1.1: pharmacological information). Volunteers who presented psychiatric comorbidities such as: psychotic signs and symptoms, neurological diseases, epilepsy, mental retardation, as well as patients with obsessive-compulsive disorders, severe personality disorders, or who had a diagnosis of heart problems were excluded. All individuals were invited to clarify their doubts after signing the consent form. Table **1** contains the descriptive characteristics of the sample with intergroup and general comparisons, anthropometric, body composition, physiological, and performance variables (Fig. **2**).

2.3. Primary Outcome Variables

HRV variables in the time and frequency domains, as well as affective responses, were designated as primary outcomes. The HRV variables analyzed included: mean HR: mean heart rate (bpm); SDNN: Standard deviation of NN intervals, a global measure of HRV (ms); RMSSD: Root mean square of the squares of successive differences between adjacent NN intervals (ms); Low frequency (LF) – component that varies between 0.04 and 0.15 Hz, corresponding to the joint action of the vagal and sympathetic components, however, with sympathetic predominance; High frequency (HF) –component that varies between 0.15 and 0.4 Hz, corresponding to vagal action; The LF/HF ratio - baroreceptor band (UN). Variables related to affect were determined by the level of positive, neutral, or negative sensation provided by aerobic exercise (−5 to +5).

2.4. Secondary Outcome Variables

Secondary outcomes included variables from the SUDS anxiety scale, which was represented by a Likert-type scale ranging from 0 (no anxiety) to 10 points (extreme anxiety), as well as the presentation of the Cartesian plane represented by the circumplex model of affect.

2.5. Potential Confounders and Effect Modifiers

The use of drugs to control BD symptoms can be considered a potential confounding factor and should be carefully considered. Medications, such as Lithium, whether or not associated with other types of drugs, can influence both the physical capacity of participants and the affective response to exercise. These effects can mask the effectiveness of the intervention, making it difficult to attribute changes in results exclusively to exercise, thus reducing confidence in the estimate of effect. However, it is worth noting that in studies of this nature, it is unlikely that physicians and professionals in the field will eliminate the use of such drugs, since non-use can cause deleterious effects and suffering to the patient.

3. PROCEDURES

3.1. Anthropometry and Baseline Measurements

Skinfold thickness measurements were taken using the three-fold protocol (chest, abdomen, and thigh) as described by Jackson and Pollock [18], as well as waist, abdomen, and hip circumference measurements. After predicting body density, fat percentage was estimated using the Siri equation [19]. Data were also collected on resting heart rate (HRrest - Polar® 810i), and resting blood pressure (BP) using a digital sphygmomanometer (WanMed®), in addition to body mass and height (Sanny®, Brazil).

3.2. Submaximal Incremental Test for VO_2Max Estimation

In a calm and temperature-controlled environment (20 to 22°C), all participants had their resting HR measured for 6 min before the test. The warm-up consisted of walking on a treadmill at 5.0 km·h^{-1} at 1% incline for 3 min. From this initial stage onwards, 2% increases in the incline (approx. 1 MET) were made every minute with the aim of reaching a

Table 1. Comparison of baseline measures and sample characteristics. Measures described by mean ± SD, as well as, ($CI_{95\%}$).

General Parameters	Control (n=8)	Bipolar (n=8)	General (n=16)	*p* value
Age (years)	42.4 ± 10.9 (33.7 – 51.0)	46.6 ± 10.4 (33.9 – 55.2)	44.5 ± 10.2 (39.0 – 49.9)	0.426
Body mass (kg)	78.0 ± 16.0 (64.6 – 91.3)	85.5 ± 16.8 (71.4 – 99.5)	81.8 ± 16.3 (73.0 – 90.0)	0.377
Height (cm)	161.4 ± 11.2 (151.9 – 170.7)	166.3 ± 8.3 (159.4 – 173.2)	163.9 ± 9.9 (158.6 – 169.1)	0.330
Fat (%)	27.0 ± 10.7 (18.0 – 35.8)	25.8 ± 11.3 (16.3 – 35.1	26.4 ± 10.6 (20.7 – 32.0)	0.832
Abdominal (cm)	95.2 ± 11.9 (85.2 – 105.1)	102.8 ± 17.2 (88.4 – 117.1)	99.0 ± 14.8 (91.0 – 106.9)	0.720
Waist hip ratio (cm/cm)	0.85 ± 0.08 (0.77 – 0.91)	0.88 ± 0.09 (0.80 – 0.95)	0.87 ± 0.09 (0.81 – 0.91)	0.419
Physiological Variables				
HR_{rest} (bpm)	81.1 ± 11.1 (71.8 – 90.3)	81.0 ± 10.4 (72.3 – 89.6)	81.1 ± 10.4 (75.5 – 86.5)	0.982
HR_{max} (bpm)	177.6 ± 10.4 (168.9 – 186.2)	173.4 ± 10.4 (164.7 – 182.0)	175.5 ± 10.2 (170.0 – 180.9)	0.426
Systolic BP (mmHg)	124.3 ± 12.4 (113.9 – 134.5)	130.8 ± 14.4 (118.7 – 142.9)	127.6 ± 13.4 (120.4 – 134.7)	0.341
Diastolic BP (mmHg)	81.8 ± 15.2 (69.0 – 94.4)	82.0 ± 9.7 (73.8 – 90.1)	81.9 ± 12.3 (75.3 – 88.4)	0.969
VO_2Max ($mL{\cdot}kg^{-1}min^{-1}$)	31.3 ± 2.9 (28.8 – 33.7)	26.2 ± 7.6 (19.8 – 32.5)	28.8 ± 6.2 (25.4 – 32.0)	0.110
METs	9.0 ± 0.8 (8.2 – 9.6)	7.5 ± 2.2 (5.6 – 9.3)	8.2 ± 1.8 (7.2 – 9.1)	0.110
Psychiatric Scales				
Hamilton Depression (score) [16]	-	5.3 ± 4.4		
Young Mania (score) [17]	-	1.0 ± 1.4		

Abbreviations: SD = standard deviation; HR = heart rate; B*P* = blood pressure; METs = metabolic equivalent; VO_2Max = maximum oxygen consumption; Abdominal = Abdominal circumference.

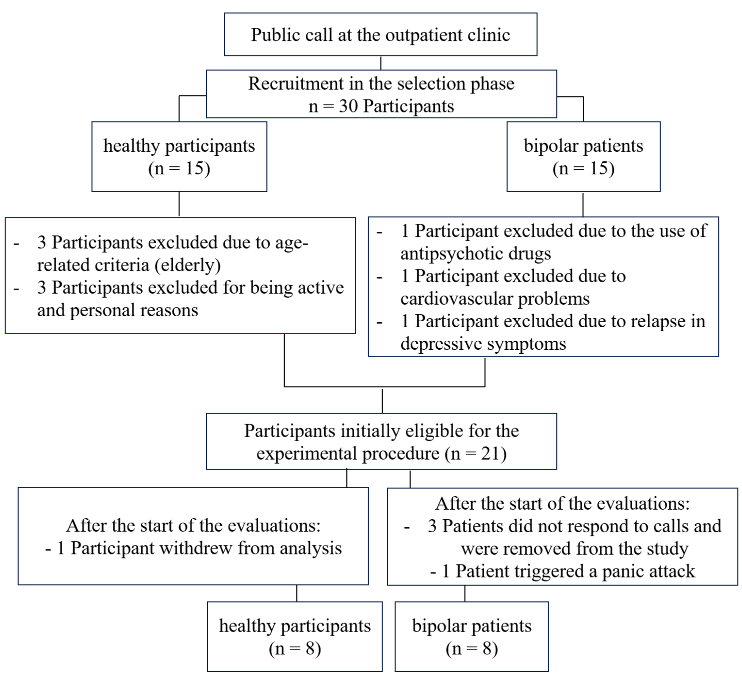

Fig. (2). Inclusion and exclusion criteria for the sample participants. (*A higher resolution/colour version of this figure is available in the electronic copy of the article*).

minimum intensity of 55% of the HR reserve (RHR). After reaching the proposed intensity, the incline and speed were kept unchanged for 6 min to allow the achievement of a steady state. The aim was to reach a minimum of 65% of the RHR. HR and rate perceived exertion (RPE), ranging from 0 (no exertion) to 10 (absolute maximum), were recorded in the final 10 seconds of each minute. The mean HR between the 5th and 6th minutes of the steady state phase was used to estimate VO_2Max. Maximal heart rate (HRmax) was estimated using the predictive equation 220 minus age. VO_2Max was estimated from the walking equation proposed by ACSM [20], as adapted by Oliveira *et al*. [21] or treadmills, together with the predictive equation defined by Swain *et al*. [22]. The reliability of the measurement for treadmills was previously established by Santos *et al*. [23]. Table **2** presents the VO_2Max prediction equation.

Table 2. VO_2Max prediction proposed by Swain *et al*., and adapted by Oliveira *et al*. for depressed patients.

$$VO_2Max = [(0.1 \cdot \text{velocity}) + (1.8 \cdot \text{velocity} \cdot \text{slope}) + 3.5] \div [(HR_{Workload} - HR_{Rest}) \div (HR_{Max} - HR_{Rest})] + 3.5$$

Where:

VO_2Max - maximum oxygen consumption in $mL \cdot kg^{-1} \cdot min^{-1}$;

Velocity - $m \cdot min^{-1}$;

slope - in centesimal values;

$HR_{Workload}$ - mean HR between the 5th and 6th min in steady state;

HR_{Rest} - HR after 10 minutes of rest;

HR_{Max} – estimated using equation 220 − age.

3.3. Heart Rate Variability (HRV)

HRV data were collected at two time points: immediately before and after each exercise session (continuous or HIIT). On both occasions, participants remained seated in a reclining chair, in a silent, dark room with controlled temperature (21–23°C), breathing spontaneously and with their eyes blindfolded to minimize external sensory stimuli. The heart rate monitoring strap (RS800 Precision Performance version 4.01.029, Polar, Finland) was positioned over the mesosternal region in a standardized manner to ensure accurate detection of cardiac signals. Recordings were obtained during 10 minutes of absolute rest, with participants in a fully relaxed physical and mental state. It is important to note that the standardization of respiratory cycles using a metronome was not implemented due to the concurrent acquisition of EEG biological signals, as any external noise could potentially interfere with signal quality (secondary data). HRV data were transmitted to the computer to be recorded in specific software. The stored HRV of each participant was imported into a software capable of calculating HRV measurements (Kubios HRV – Heart Rate Variability Analysis – Version 3.0.2, 2017), and analyzed in the time and frequency domain, extracting the following derived data: SDNN (standard deviation of normal-to-normal RR intervals); RMSSD (square root of the differences of successive RR intervals); VLF (very low frequency band - .003 - 0.04 Hz); LF (low frequency band - 0.04-0.15 Hz); HF (high frequency band - 0.15-0.4 Hz); and LF/HF ratio (ms^2).

3.4. Psychometric Scales and Rate of Perceived Exertion

Felt arousal, feeling scales, and anxiety SUDs scale were applied at two times: pre-exercise and five min after exercise. The RPE (Borg 0-10) was applied during and post 15 min of the exercise session.

3.3.1. Felt Arousal Scale

The level of bodily arousal was analyzed in a self-perceived manner during the experimental conditions: pre- and post-exercise. It is a Likert scale, which varies linearly from 1 = slightly aroused to 6 = very aroused, with intermediate values.

3.3.2. Feeling scale

The dimensions of affective responses are determined by the level of positive, neutral, or negative sensation provided by aerobic exercise, being distributed on a bipolar ordinal scale, ranging from zero (0) as a neutral position; +1 = reasonably good to +5 = very good; -1 = reasonably bad, up to -5 = very bad.

3.3.3. SUDS Anxiety Scale

Each participant described how they feel at the moment according to their perceived mental state, scoring on a linear scale between 0- absolutely no anxiety, and 10- extremely anxious.

3.3.4. RPE

The adapted linear RPE scale (0 to 10) as produced and described by Borg was used, where “0” refers to the perception of ‘extremely light’ effort, reaching “total fatigue” at 10. For the self-selected sessions, participants were induced and motivated to obtain the maximum global effort index.

3.3.5. Construction of the Circumplex Model of Affect

Affect was assessed using the circumplex model, a two-dimensional structure that positions the values of psychological processes obtained from emotional responses of arousal (activation) and affective (pleasantness). Then, based on these scales of sensations and activation, a representative Cartesian model was created with four divisions that demonstrate whether the subject is in a quadrant of emotional categories related to serenity, calm, comfort, relaxation and drowsiness (in case of low arousal and positive affective value) or in an emotional category of joy, happiness, enchantment, surprise and stimulation (high arousal and positive affect). When the level of arousal is high, and the affective response is negative, the emotional characteristics are related to a furious, distressed, irritated, frustrated, tense, fearful, excited, and panicked state. Finally, when the level of arousal is low and the affective response is negative, the emotional states are characterized as sad, depressed, melancholic, bored, dejected, and tired.

3.4. Continuous Aerobic and HIIT Session

Both continuous and HIIT procedures were initiated with a standard warm-up of 5 min walking at 40% VO_2Max (no incline). For the continuous exercise, participants performed 12 minutes at 65% VO_2Max. To maintain the proposed intensity, participants walked under the influence of high percentages of incline, maintaining a constant speed of 5.0 $km \cdot h^{-1}$. Subsequently, a 3-minute cool-down lap was performed at free intensity.

For the HIIT exercise, participants performed a total of 6 stimuli of 45 seconds, with a recovery of 1 minute and 15 seconds between stimuli. The intensity of the exercise bouts was set at a level corresponding to 100% of the external load associated with VO_2max, with recovery at at 40% of the external load corresponding to VO_2max, maintaining a constant speed of 5.0 $km \cdot h^{-1}$. Active recovery at 40% of the external load corresponding to VO_2max was chosen based on evidence indicating that recovery at intensities below 50% of the load associated with VO_2max enhances the ability to sustain exercise intensity during subsequent bouts of high-intensity efforts. To estimate the intensity corresponding to 40% of the external load associated with VO_2max, the variables mentioned above were applied to the equation (Table **2**), which yielded the estimated intensity corresponding to 40% of the external load associated with VO_2max [24-29]. At the end, a calm lap of free intensity was established for 3 minutes. The total exercise time for both groups was 20 minutes, and the total work was equal between the groups. The intensities for both continuous and HIIT were defined and adjusted by metabolic conversion, using as a reference the VO_2Max previously estimated from the prediction equations for running and walking described by ACSM [20] [(velocity = $VO_2Max - 3.5 \div 0.2$) $m \cdot min^{-1}$]. The criterion for completing the protocol was the achievement of all stimuli or maximum voluntary exhaustion. HR and RPE were recorded during all training sessions. A professional treadmill (ATL, Inbramed®, Brazil) was used to carry out the exercise program.

It is worth noting that the interval adopted between exercise sessions (HIIT and Continuous aerobic) was at least 72 hours and at most 96 hours, which is consistent with the literature to date. A recent and elegant systematic review [30], which investigated the duration of the effect of a high-intensity interval training session on muscle damage markers, revealed that muscle damage indicators remain elevated only for up to 48 hours. Therefore, 72 hours, as proposed in the present study, characterizes the washout period.

3.5. Training Impulse Calculation (TRIMP)

Once the workload has been managed from incremental slope percentages, we convert the estimated metabolic demand into metabolic equivalents (METs). Then, to calculate the TRIMP, as suggested by Banister and adapted by Foster, the METs value was multiplied by the training HR [31,32].

3.6. Randomization Process

Participants, after eligibility and written informed consent, were randomly assigned to experimental assessments of continuous or HIIT. A simple randomization was used. The randomization sequence was computer-generated (www.randomizer.org) and allocated by a third member of the research group, keeping it blind until the time of the experiment.

3.7. Data analysis, Processing, and Bias

To avoid possible biases in the analysis, the data were collected by a researcher associated with the project and the research group (A.S.F. and E.L.) and analyzed by a third researcher (group leader S.M.). The researcher responsible for data analysis remained blind throughout the data collection process. The names of all participants remained confidential, being excluded from the technical file and replaced by numbers.

3.8. Possible Unexpected Effects

Unexpected or adverse effects are commonly attributed to physical exercise. In addition, high-intensity interval training is commonly the target of painful results or significant acute deleterious effects. Such effects were managed and, where necessary, appropriate intervention was undertaken. Deleterious outcomes are presented in the results.

3.9. Size Study

Using the G-Power statistical package (Free Version 3.0.5) for analysis, "ANOVA repeated measures within factors", with two measures and two groups, the sample size considered an error of 5%, statistical power of 80%, and an effect size of 0.40, resulting in 16 participants. Considering the sample loss and significant dropout in the conduct of chronic experimental models, a larger number of participants was recruited, expecting a loss of around 20% to 30%.

3.10. Statistical Analysis

A descriptive analysis of the data was performed and presented as mean ± standard deviation (SD), 95% confidence interval ($CI_{95\%}$), or median values. Data normality was tested using the Shapiro-Wilk test. If normality was present ($p > 0.05$), data were expressed as mean ± standard deviation. If normality was violated ($p < 0.05$), data were expressed as median, maximum, and minimum values. To test differences between groups for baseline anthropometric and physiological variables, a Student's t-test for independent samples was used. The mixed two-way analysis of variance (ANOVA) followed by the Bonferroni post-hoc test was used to determine which specific groups exhibit statistically significant differences, after the ANOVA test indicated that there is an overall difference between the means of the groups under study, as well as the effect size (ES) from the partial Eta (η^2p) (METs, $HR_{Workload}$, $\%HR_{Workload}$, TRIMP).

For the analyses of HRV variables, the Wilcoxon test compared the pre- and post-intervention conditions of measurements in the time domain (SDNN, RMSSD) and frequency (VLF, LF, HF, and LF/HF ratio). For comparison between groups, the Mann-Whitney test was used. The same occurred for the felt arousal, feeling, and anxiety scales, which were treated and compared in a non-parametric manner. The magnitude of the ES for nonparametric tests ($r = z \div \sqrt{n}$) was determined to support the analyzed results. All analyses were performed using SPSS (v. 20, SPSS Inc., Chicago, USA), and the graphs were produced using GraphPad Prism software (version 8). A significance level of $p = 0.05$ was considered.

3. RESULTS

3.1. General Analytical Information

After Shapiro-Wilk distribution analysis, normality was observed for all baseline variables, respectively for control group and bipolar patients: Age ($p = 0.768$; $p = 0.252$); Body mass ($p = 0.441$; $p = 0.449$), Height ($p = 0.533$; 0.519), Abdominal circumference ($p = 0.140$; $p = 0.550$), fat percentage ($p = 0.303$; $p = 0.614$), Waist-to-hip ratio ($p = 0.174$; $p = 0.618$), SBP ($p = 0.755$; $p = 0.099$), DBP ($p = 0.187$; $p = 0.761$), HR_{Rest} ($p = 0.631$; $p = 0.543$), HR_{Max} ($p = 0.768$; 0.252), VO_2Max ($p = 0.149$; $p = 0.245$), METs HIIT protocol ($p = 0.115$; $p = 0.253$), METs continuous protocol ($p = 0.137$; $p = 0.261$), TRIMP HIIT ($p = 0.512$; $p = 0.144$), TRIMP Continuous ($p = 0.073$; $p = 0.156$). Additionally, the ordinal variables observed in the felt arousal, feeling, and anxiety SUDs scales were automatically treated in a non-parametric way.

3.2. Pharmacological Information

100% of BD patients reported using Lithium (300 mg doses). Of these 100%, 37.5% associated Lithium with Rivotril (2 mg), 25% associated Lithium with Lamotrigine (100 mg), 25% associated Lithium with Diazepam, and only one patient reported using Fenargan (25 mg) in combination.

In detail, all eight patients diagnosed with BD were receiving pharmacological treatment. Lithium was the most frequently prescribed medication, administered to all participants, generally at a dose of 300 mg, taken two to three times per day. Additionally, several patients received adjunctive psychotropic medications: clonazepam (Rivotril, 2 mg at night) was prescribed to patients 1, 3, and 6; lamotrigine (100 mg) was used by patients 2 and 4; patient 2 reported taking 1.5 tablets at night, while patient 4 took 2 tablets at night. Diazepam was administered to patients 5 and 7, at doses of 20 mg and 10 mg at night, respectively. Patient 2 also used promethazine (Phenergan, 25 mg), with 0.5 tablets in the morning and 1 tablet in the evening. Patient 3 received cyamemazine (Neuleptil, 10 mg at night), and patient 8 was treated with sertraline (50 mg at night). Patients 5 and 7 were unable to precisely report their lithium dosing regimens.

3.3. Exercise Parameters

After analyzing statistical assumptions, the two-way mixed ANOVA showed an effect for the METs variable and the types of interventions ($F_{(1,14)} = 128.04$; $p < 0.001$; $\eta^2p = 0.901$), but not between groups ($F_{(1,14)} = 3.176$; $p = 0.096$; $\eta^2p = 0.185$). There was no interaction between group and intervention factors ($F_{(1,14)} = 2.652$; $p = 0.126$; $\eta^2p = 0.159$). For the $HR_{Workload}$ variable, the two-way ANOVA showed a similar result, differing significantly between interventions ($F_{(1,14)} = 8.868$; $p = 0.010$; $\eta^2p = 0.388$), but not between groups ($F_{(1,14)} = 0.681$; $p = 0.423$; $\eta^2p = 0.046$), and showing no interactions between group and intervention ($F_{(1,14)} = 0.181$; $p = 0.677$; 0.013). As there were differences in the $HR_{Workload}$ percentage (%) achieved during the HIIT and continuous exercise interventions, there were also differences between the percentages observed between interventions ($F_{(1,14)} = 8.785$; $p = 0.010$; $\eta^2p = 0.386$), with no differences between groups ($F_{(1,14)} = 3.632$; $p = 0.077$; $\eta^2p = 0.206$) or group x intervention interactions ($F_{(1,14)} = 0.260$; $p = 0.618$; $\eta^2p = 0.018$). Finally, following the same line of results, the TRIMP variable differed between interventions ($F_{(1,14)} = 93.836$; $p < 0.001$; $\eta^2p = 0.870$), however, not between groups ($F_{(1,14)} = 2.124$; $p = 0.167$; $\eta^2p = 0.132$), or for interactions ($F_{(1,14)} = 1.601$; $p = 0.226$; $\eta^2p = 0.103$), suggesting a greater physiological impact derived from the HIIT intervention. The exercise parameters are presented in Table **3**.

Effect size analysis comparing the HIIT and continuous exercise conditions in the control group revealed large magnitudes across most physiological parameters. A very large effect was observed for the METs variable. Regarding $HR_{Workload}$ responses (bpm), presented a moderate-to-large effect size, while $HR_{Workload}$ (%) showed a moderate effect. Finally, TRIMP exhibited a very large effect size, suggesting a substantially greater physiological load associated with the HIIT condition compared to continuous exercise in the control group.

4. PRIMARY OUTCOME

4.1. Heart Rate Variability (HRV)

The HRV results, expressed as mean ± SD, are presented in Table **4**. Due to the different normality results between conditions (time · group · intervention), as well as the sample size, the data were analyzed nonparametrically. The Wilcoxon test determined the intra-group differences, with subsequent Mann-Whitney analysis to determine the effects between groups. The analyses were divided between domains (time and frequency).

4.2. HRV in the Time Domain

Regarding RMSSD, both groups (control and BD) and sessions (HIIT or continuous exercise) demonstrated a significant increase in this indicator (Table **1**). Conversely, with respect to SDNN, a significant elevation in this parameter was observed exclusively among individuals diagnosed with BD across both exercise modalities (HIIT and continuous exercise).

When the Mann-Whitney test was applied, no differences were observed between groups in the pre or post exercise conditions, both for continuous (SDNN: U = 13.000, $p = 0.05$; U = 26.000, $p = 0.529$; RMSSD: U = 30.500, $p = 0.878$; U = 24.000, $p = 0.442$, respectively for the pre *vs.* pre and post *vs.* post comparison), and for HIIT (SDNN: U = 24.500, $p = 0.442$; U = 31.000, $p = 0.959$; RMSSD: U = 26.000, $p = 0.529$; U = 30.000, $p = 0.834$, respectively for the pre *vs.* pre and post *vs.* post comparison). For illustration purposes only, the pre- and post-exercise, HIIT, and continuous percentage deltas, as well as their respective effect sizes, were reported in Fig. (**3**).

Table 3. Exercise and performance setup parameters.

Parameters	HIIT					
	Control (n=8)			Bipolar (n=8)		
	Mean	SD	$CI_{95\%}$	Mean	SD	$CI_{95\%}$
Slope (%)	13.0	1.9	(11.4 - 14.6)	9.6	5.1	(5.3 - 17.2)
Demand (mL·kg^{-1}min^{-1})	31.3	2.9	(28.8 - 33.7)	26.2	7.6	(19.8 - 32.5)
METs	**9.0***	0.8	(8.2 - 9.6)	7.5	2.2	(5.6 - 9.3)
$HR_{Workload}$ (bpm)	**156.1***	9.3	(156.1 - 163.8)	159.6	9.0	(152.1 - 167.1)
$HR_{Workload}$ (%)	**88.0***	5.1	(83.7 - 92.2)	92.1	3.2	(89.4 - 94.8)
TRIMP (METs · bpm)	**1399.5***	176.7	(1251 - 1547)	1196.0	357.4	(897 - 1494)
Continuous Exercise						
Slope (%)	8.5	1.3	(7.4 - 9.5)	6.2	3.3	(3.5 - 9.0)
Demand (mL·kg^{-1}min^{-1})	24.5	1.9	(22.9 - 26.1)	21.2	5.0	(17.0 - 25.3)
METs	**7.0***	0.5	(6.5 - 7.5)	6.1	1.4	(4.8 - 7.2)
$HR_{Workload}$ (bpm)	**150.5***	3.6	(147.4 - 153.5)	152.1	7.2	(146.1 - 158.1)
$HR_{Workload}$ (%)	**85.0***	5.6	(80.2 - 89.6)	87.8	3.2	(85.1 - 90.4)
TRIMP (METs · bpm)	**1053.5***	179.3	(965 - 1116)	920.3	218.9	(737 - 1103)

Abbreviations: Demand = Metabolic demand in the workload performed; METs = Metabolic equivalent unit — One MET is equivalent to a person's oxygen consumption at rest, which is approximately 3.5 mL of O_2 per kilogram of body weight per minute; $HR_{Workload}$ = Absolute and percentage heart rate achieved throughout the exercise; TRIM*P* = training impulse: product of metabolic equivalent multiplied by HR; * = significant differences between types of interventions.

Table 4. HRV results in the time and frequency domains before and after exercise interventions.

Parameters	HIIT				Continuous Exercise			
	Control		Bipolar		Control		Bipolar	
	Pre	Post	Pre	Post	Pre	Post	Pre	Post
SDNN (ms)	22.5 (15.9)	32.2 (20.3)	13.8 (7.6)	**34.4 (27.5)***	29.3 (18.4)	31.5 (16.6)	17.2 (18.6)	**30.6 (30.8)***
RMSSD (ms)	22.4 (20.7)	**36.3 (33.2)***	11.6 (7.4)	**37.1 (31.7)***	25.9 (25.5)	**36.4 (27.1)***	19.3 (24.5)	**29.1 (26.8)***
VLF (ms^2)	3.61 (0.8)	2.77 (1.3)	3.16 (1.0)	**1.70 (0.8)***	3.59 (0.6)	**2.70 (1.0)***	3.41 (1.3)	**2.10 (0.6)***
LF (ms^2)	5.50 (1.1)	4.56 (1.9)	5.42 (1.4)	**3.75 (1.3)***	5.60 (0.7)	4.22 (1.7)	5.29 (1.6)	**3.80 (0.9)***
HF (ms^2)	5.17 (2.3)	**4.06 (2.3)***	5.28 (1.8)	**2.89 (1.8)***	5.37 (1.4)	**3.61 (1.8)***	5.12 (1.9)	**3.55 (1.9)***
LF/HF Ratio	1.50 (1.2)	1.53 (1.0)	1.17 (0.7)	**1.97 (1.6)***	1.92 (1.7)	2.78 (3.1)	2.01 (2.5)	2.32 (2.4)

Data are presented as mean values followed by standard deviation in parentheses. Pre- and post-exercise values are shown for both the high-intensity interval training (HIIT) and continuous exercise protocols in control participants and individuals with bipolar disorder. SDNN: standard deviation of NN intervals; RMSSD: root mean square of successive differences; VLF: very low frequency; LF: low frequency; HF: high frequency; LF/HF: interaction between baroreflex mechanisms and overall autonomic modulation. *Indicates significant difference between pre- and post-exercise within the same group ($p < 0.05$).

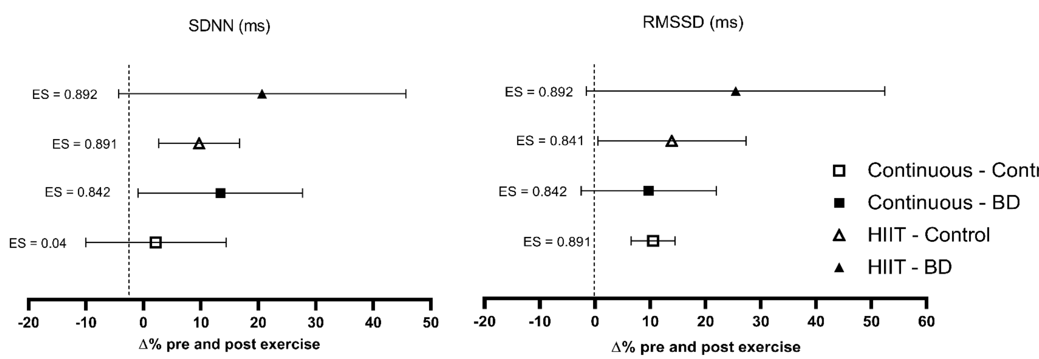

Fig. (3). The HRV results in the time domain. Δ% = percentage change pre and post intervention; Continuous Control and Continuous BD = moderate-intensity continuous exercise condition in control participants and patients with bipolar disorder, respectively; HIIT Control and HIIT BD = high-intensity interval exercise condition in control participants and patients with bipolar disorder, respectively. (*A higher resolution/colour version of this figure is available in the electronic copy of the article*).

4.3. HRV in the Frequency Domain

Regarding frequency domain variables, the group of patients with BD significantly increased VLF and LF values and reduced HF values for both exercise sessions (HIIT and continuous exercise). Meanwhile, the LF/HF ratio showed a significant increase only in the HIIT session (Table **1**). Whereas, for the control group, the Wilcoxon test revealed only a significant reduction for the HF index in both sessions (HIIT and continuous) and an increase in the LF/HF ratio only in the HIIT session (Table **1**). When the Mann-Whitney test was applied for analysis between groups, no significant differences were observed for VLF in continuous exercise (Pre: U = 23.000, p = 0.382; Post: 22.500, p = 0.318) and for HIIT exercise (Pre: U = 18.000, p = 0.141; Post: U = 17,000, p = 0.130). The same occurred for the LF variable in continuous exercise (Pre: U = 24.500, p = 0.431; Post: U = 24.500, p = 0.434) and HIIT exercise (Pre: U = 28.000, p = 0.674; Post: 23.000, p = 0.345) and for the HF variable in continuous exercise (Pre: U = 27.000, p = 0.600; Post: U = 31.500, p = 0.958) and in HIIT exercise (Pre: U = 31.000, p = 0.916; Post: U = 22.000, p = 0.294). For illustration purposes only, the pre- and post-exercise, HIIT, and continuous percentage deltas, as well as their respective effect sizes, were reported in Fig. (**4**).

4.4. Felt Arousal and Feeling Scale Responses

The felt arousal and feeling scales were treated and analyzed nonparametrically. The Wilcoxon test revealed significant differences pre and post continuous exercise (control: Z = −2.598, p = 0.009; BD: Z = −2.232, p = 0.026) and HIIT (control: Z = −2.539, p = 0.011; BD: Z = −2.552, p = 0.011) for the activation scale, suggesting an increase in body arousal. When analyzing the differences between groups, the Mann-Whitney test did not indicate significant differences between control and BD participants for the pre (continuous: U = 18.000, p = 0.113; HIIT: U = 27.000, p = 0.584) and post (continuous: U = 14.500, p = 0.056; HIIT: 31.500, p = 0.954) conditions, demonstrating that both exercise conditions led to the same level of arousal.

For the feeling scale variable, the Wilcoxon test demonstrated differences between the pre and post exercise conditions (control: Z = −2.588, p = 0.010, ES = -0.90; BD: Z = −2.032, p = 0.042, ES = -0.72) and HIIT (control: Z = −2.555, p = 0.011, ES = -0.90; BD: Z = −2.555, p = 0.011, ES = -0.92). Between groups, there were no significant differences on the affective responses for the controlparticipants (Pre: U = 28,000, p = 0.656; Post: U = 26,000, p = 0.511) and participants with BD (Pre: U = 29,000, p = 0.701; Post: U = 23,000, p = 0.214), demonstrating that both exercises induced positive responses to the proposed configurations. In addition, the magnitude of the change in the feeling scale in relation to HIIT training was greater than in the continuous condition, as observed in Fig. (**5**).

5. SECONDARY OUTCOME

5.1. Circumplex Model and its Relationship with Affective Valence

Affect can be described and understood through the Cartesian circumplex model. Its dimensions are bipolar and orthogonal and are named based on the pleasure or displeasure experienced and arousal perceived as high or low. Fig. (**6**) presents the effects of the different types of exercises in relation to the quadrants plotted in the circumplex model.

5.2. Anxiety Responses to Exercise

The Wilcoxon test showed differences in the pre and post comparison for control participants in the face of continuous training (Pre: 1.13 ± 0.83, Post: 0.25 ± 0.46; Z = −2.133, p = 0.020; ES = -0.63), but the same did not occur with the HIIT intervention (Pre: 1.13 ± 1.36, Post: 0.38 ± 0.74; Z = −1.857, p = 0.063). This result can be explained by the great variability of response between participants, as observed in Fig. (**7**). Patients with BD appear to have benefited more widely from the continuous (Pre: 3.63 ± 1.19, Post: 2.00 ± 1.51; ES = -0.50) and HIIT (Pre: 3.38 ± 1.41, Post: 0.88 ± 0.99; ES = -0.85) interventions, presenting pre and post differences for both conditions (Z = −1.414, p = 0.016 and Z = −2.410, p = 0.016, respectively for continuous and HIIT), and with a greater magnitude of the effect observed.

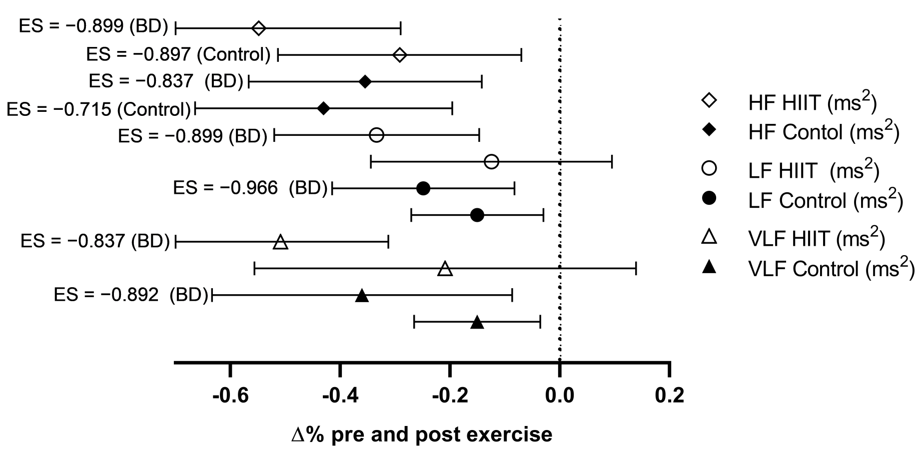

Fig. (4). The HRV results in the frequency domain. Δ% = percentage variation pre and post intervention; BD = patients with bipolar disorder. (*A higher resolution/colour version of this figure is available in the electronic copy of the article*).

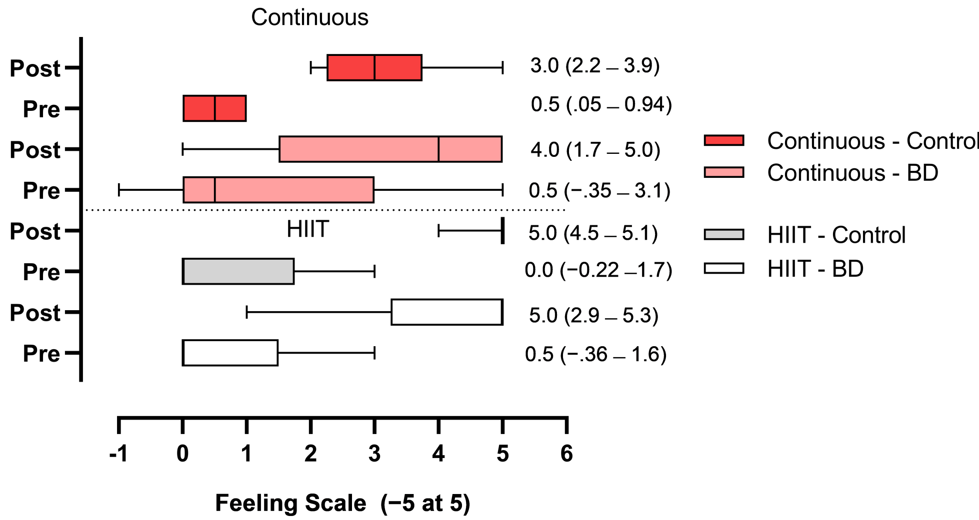

Fig. (5). Effects of exercise on the feeling scale. Continuous Control = effects of continuous intervention in control participants; Continuous BD = effects of continuous intervention in participants with bipolar disorder; HIIT Control = effects of high-intensity interval intervention in control participants; HIIT BD = effects of high-intensity interval intervention in participants with bipolar disorder; values are described by median and CI$^{95\%}$. (*A higher resolution/colour version of this figure is available in the electronic copy of the article*).

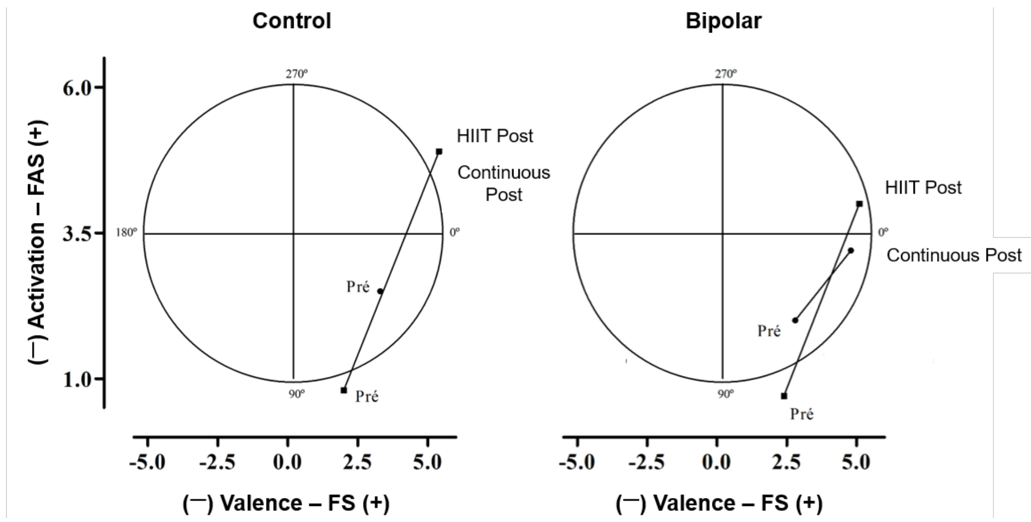

Fig. (6). Observational cirumplex model of behavior in the face of Continuous and HIIT exercises. On the right, we can see the cartographic model for the BD group. On the left, we can see the cartographic model for the control group. FAS = felt arousal scale; FS= feelings scale. (*A higher resolution/colour version of this figure is available in the electronic copy of the article*).

When applied to comparison between groups, the Mann-Whitney test showed differences from pre-intervention entry for control (1.1 ± 1.3) and BD (3.6 ± 1.1) participants to the continuous intervention (Pre: U = 3.000, p = 0.001). The same difference between control (1.1 ± 1.3) and bipolar (3.4 ± 1.4) participants was observed for the HIIT condition (Pre: U = 8,000, p = 0.010). These results denote a potential chronic anxiogenic condition related to BD. The post-exercise effects showed differences between groups for the continuous condition (control: 0.3 ± 0.4 *vs.* BD: 2.0 ± 1.5; U = 7,000, p = 0.006), but not for the HIIT intervention (control: 0.4 ± 0.7 *vs.* BD: 0.9 ± 0.9; U = 21,000, p = 0.195).

5.3. Correlational Models between Dependent Variables

A Spearman correlation was performed based on the differences between the pre and post conditions and the TRIMP performance values. No correlations were observed between the feeling scale and the TRIMP values, either for the continuous condition (p = 0.171) or for the HIIT condition (p = 0.159) in controlparticipants, as well as for participants with BD (p = 0.123; p = 0.237, respectively, for continuous exercise and HIIT).

6. DISCUSSION

To our knowledge, this study appears to be a pioneer in examining the acute effects of moderate-intensity continuous aerobic exercise *vs.* HIIT on autonomic and psychoaffective parameters in patients with BD. First, the HIIT exercise protocol is established as the highlight of our investigation, since the literature basically presents interventions with continuous and moderate characteristics in the context of mental health. Therefore, establishing the feasibility of an

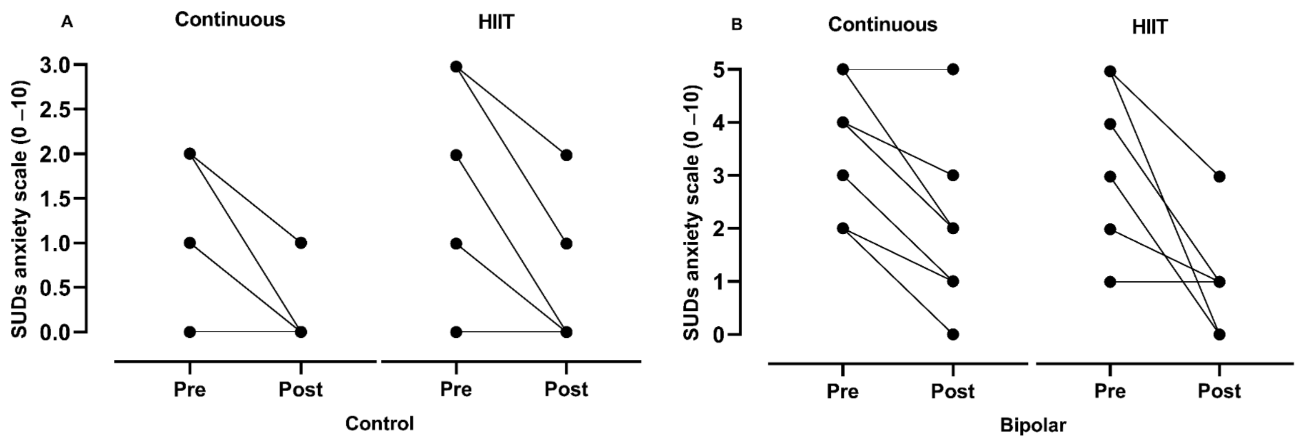

Fig. (7). Responders and non-responders to continuous and HIIT exercise on anxiety responses. (**A**) presents the responses of the control group; (**B**) presents the responses of the group with BD. (*A higher resolution/colour version of this figure is available in the electronic copy of the article*).

interval exercise configuration, which despite the intensity administered (100% of VO_2Max), was capable of fractionating the physiological impact and inducing affective responses favorable to adherence to regular exercise, provides important support for Physical Education professionals and Physiotherapists to act in their rehabilitation and physical reconditioning routines, in addition to filling a relevant gap that is still unknown.

Thus, as the main hypothesis (H1), we believed that the variables related to HRV would present differences before and after the intervention and between control participants and BD individuals. Our hypothesis was partially accepted, since there were no differences between groups only for HRV measurement under the time and frequency domains. Furthermore, FAS and FS were also positively modified after exercise, but there were no differences between groups, suggesting equality between procedures. Furthermore, positive affective responses for both groups (control and BD) and both exercise models were observed, however, with superior responses for HIIT exercise.

6.1. General Information

Although traditional aerobic endurance exercises have well-documented benefits for both apparently healthy individuals [33, 34] and patients with BD [2, 3, 35-38], recent literature seeks to deepen our understanding of the optimal exercise intensity for health improvement and to further elucidate the role of maximum aerobic power (VO_2Max) as a critical marker of cardiovascular mortality risk [39, 40]. So, when we compare lower intensity protocols to HIIT, in healthy people, the results are promising in supporting the use of exercise intensities close to or higher than VO_2Max, verifying significantly higher increases mainly for VO_2Max, and in a shorter work time (time-efficient strategy) [41-45]. Moreover, such improvement effects appear to exhibit similar behaviors among different levels of physical fitness, but there is no evidence in the literature that provides us with a basis regarding the application of the HIIT model and its effects with BD, making direct comparisons unfeasible.

A second point in question in our study was the proposition of a reasonably precise means of control for prescribing aerobic exercise, since methodologically studies carried out with bipolar patients are relatively questionable [46-49]. Therefore, the standardization of stimuli, whether continuous or HIIT, also allows us to obtain standardized responses from the perspective of science and clinical practice, and predict potential long-term benefits [50]. In our study, all exercise stimuli were balanced to generate the same physiological impact, and the results, despite maximizing the differences between the selected intensities, provided a similar and attenuated perception of effort between the modalities (3.5 ± 0.5 *vs.* 4.0 ± 0.8, on a scale of 0 to 10, respectively, for continuous exercise and HIIT exercise in BD patients).

Concerning safety, a preliminary investigation examining the effects of a 12-week HIIT protocol demonstrated that this intervention is apparently safe and well-tolerated in individuals with mental disorders, while also showing potential to reduce stress, anxiety, and depressive symptoms, as well as to improve the overall psychological well-being index [12]. On the other hand, although the results of the present study indicate that high-intensity interval exercise may be beneficial for people with BD, physical exercise appears to exert stage-dependent effects in individuals with BD, necessitating careful consideration of mood state when prescribing activity. During manic episodes, vigorous exercise should be approached with caution, as patients may be prone to overexertion and impaired judgment, whereas in euthymic phases, structured physical activity is generally safe and may confer benefits for mood stabilization and cardiovascular health [51].

A third point should be highlighted. The sample in question consisted of sedentary participants, that is, individuals who exhibit significant difficulty in maintaining any workload percentages during running. Accordingly, the HIIT protocol was intentionally designed so that the variations in proposed intensities, achieved through adjustments in treadmill incline (based on individual capacity), would facilitate more moderate cardiorespiratory responses until the required level of exertion was reached. Evidence suggests that HIIT performed at elevated percentages of incline can elicit responses similar to those observed during level running, in addition to being a time-efficient strategy [52].

6.2. Primary Outcome

There is a consensus in the literature that with higher aerobic fitness, individuals exhibit higher HRV at rest when compared to groups of sedentary or less trained individuals, denoting a greater influence of parasympathetic neural drive. Conversely, autonomic dysfunctions have been reported both for sedentary individuals and for patients with various mental disorders. [8, 11]. In this sense, unanimously, for both the control group and the BD group, as well as when comparing the different exercise sessions (HIIT *vs.* continuous exercise), the present study demonstrated a significant increase in the parasympathetic time-domain index (RMSSD), suggesting a positive parasympathetic modulation after both exercise sessions in the different groups (Table **1**). It is noteworthy that, especially for patients with BD, this is an interesting finding, as an increase in parasympathetic nervous activity indicates a good prognosis for mood stability, which in turn may result in fewer manic episodes and, thus, have an impact on the quality of life of these patients [8-11].

This increase can be partly explained by the gradual return of vagal activity after exercise, which had been suppressed during physical exertion [53]. Furthermore, in some cases, parasympathetic overcompensation is observed, in which vagal indices rise above pre-exercise resting values to promote homeostasis [54].

With respect to frequency domain metrics (VLF, LF, HF, and LF/HF), the HIIT session elicited significant alterations across all parameters in the group of individuals diagnosed with BD, characterized by reductions in VLF, LF, and HF components, and an elevation in the LF/HF ratio (Table **1**). Nonetheless, these findings warrant cautious interpretation due to the complexities involved in deciphering the physiological underpinnings that these indices may reflect. For instance, in the case of VLF, there remains considerable ambiguity surrounding the mechanisms governing its modulation [55]. Evidence suggests that the intrinsic cardiac nervous system contributes to VLF rhythmogenesis, while the sympathetic nervous system may modulate both the amplitude and frequency of its fluctuations [56]. Conversely, parasympathetic nervous system activity appears to play a role in VLF power, as parasympathetic blockade has been shown to nearly abolish this component [57]. In contrast, sympathetic blockade does not seem to significantly impact VLF power, and VLF activity persists in individuals with quadriplegia, who have impaired sympathetic innervation of the heart and lungs [58, 59]. In light of these considerations, it is not possible to precisely determine the extent to which the observed post-exercise changes in this variable were influenced by sympathetic or parasympathetic nervous activity.

The LF component, measured within the frequency range of 0.04-0.15 Hz, was formerly referred to as the baroreceptor band, as it predominantly reflects baroreceptor-mediated responses during resting states [60]. LF power can be generated through both parasympathetic and sympathetic branches of the autonomic nervous system, although blood pressure modulation *via* baroreceptors [58, 59, 61, 62] is thought to be primarily mediated by parasympathetic mechanisms [63], or exclusively through baroreflex function [64]. The sympathetic nervous system appears to be limited in generating oscillations above approximately 0.1 Hz, whereas the parasympathetic branch is capable of influencing cardiac rhythms at frequencies as low as 0.05 Hz. In resting conditions, therefore, LF power is considered to represent baroreflex activity rather than direct sympathetic cardiac modulation [56]. Accordingly, in the context of the current study, it is plausible to deduce that the significant post-exercise changes observed in this parameter at rest likely represent efforts to regulate blood pressure *via* baroreflex mechanisms or parasympathetic activation alone.

The HF band is indicative of parasympathetic nervous system activity and is often referred to as the respiratory band, as it corresponds to HR fluctuations associated with the respiratory cycle. These cyclical changes in heart rate are termed respiratory sinus arrhythmia; however, they may not represent a purely isolated marker of cardiac vagal tone [65]. HR increases during inhalation and decreases during exhalation. This occurs because, during inspiration, the cardiovascular control centers suppress vagal output, leading to an acceleration of heart rate, whereas during expiration, vagal activity is reinstated, resulting in a deceleration of heart rate *via* the release of acetylcholine [66].

In the present study, respiratory rhythm was not standardized due to logistical aspects of data collection. Participants remained resting after the exercise session without exposure to sensory stimuli (*i.e.*, absence of sound, low lighting, and minimal environmental noise). Moreover, because it is well-established that respiration control can significantly influence emotional variables, as rhythmic breathing may promote pronounced relaxation effects through efficient integration between the brain and the heart, mediated by the central autonomic network, which plays a vital role in modulating physiological and psychological stress, it is relevant to consider this context.

It is also important to highlight that the Guidelines for Reporting Articles on Psychiatry and Heart Rate Variability (GRAPH) recommend the use of spontaneous (uncontrolled) breathing during data acquisition [67]. Taking all of these considerations into account, the HF parameter may have been notably influenced by residual elevations in respiratory rate following the exercise bout, which may have impacted the post-exercise assessment outcomes.

Notwithstanding the aforementioned considerations, the absence of a standardized rhythmic breathing pattern represents a noteworthy limitation. Accordingly, we recommend that the data about the HF indicator be interpreted with caution, as it may have been substantially influenced by spontaneous (uncontrolled) respiratory activity.

Regarding the interpretation of LF/HF results, although the LF/HF ratio has long been interpreted as an indicator of cardiac sympathovagal balance, accumulating evidence indicates that such an interpretation is physiologically unsound. Several authors have demonstrated that both LF and HF components are predominantly influenced by parasympathetic modulation, rendering the LF/HF ratio incapable of distinguishing sympathetic from vagal contributions [63]. Furthermore, physiological and mathematical analyses have demonstrated that LF power cannot be considered a pure marker of sympathetic activity, nor can HF power be considered an exclusive representation of vagal tone, further discrediting the ratio as a measure of autonomic balance [68, 69]. In this sense, the LF/HF ratio may more accurately reflect the complex interaction between baroreflex mechanisms and overall autonomic modulation, rather than a direct sympathovagal relationship [64].

Thus, it is reasonable to infer that the elevated LF/HF ratio observed in the present study may reflect a predominant baroreflex-mediated mechanism involved in post-exercise blood pressure regulation, rather than an actual increase in sympathetic dominance over parasympathetic activity.

Given the known limitations of frequency-domain indices, particularly HF and the LF/HF ratio, and considering that we did not employ a controlled breathing pattern for the reasons previously described (data collection logistics and the potential influence of breathing rhythm on relaxation sensations, which could affect psychoaffective variables), we recommend interpreting these findings with caution. Conversely, we take this opportunity to underscore the relevance of the RMSSD index, emphasizing that, unanimously, across both the control and BD groups, as well as in the comparison between the distinct exercise modalities (HIIT *vs.* continuous exercise), the present investigation revealed a significant elevation in this index, indicating favorable parasympathetic modulation following both exercise sessions in the respective groups (Table **1**). It is worth emphasizing that, particularly for individuals with BD, this represents a noteworthy and meaningful outcome, as an enhancement in parasympathetic nervous activity is associated with a favorable prognosis for mood stabilization, which, in turn, may contribute to a reduction in manic episodes and consequently improve these patients' quality of life [8-10].

On the other hands, the concept of affective responses revolves around how the human brain perceives and provides meaning to a given external sensory stimulus, or resulting from changes in the internal physiological environment, mediating a facilitatory behavior capable of stimulating the reward systems through dopaminergic pathways, or a negative behavior, understood by the same as an environment of tension, fear, or anxiety [70, 71]. That said, the evidence available in the literature regarding the HIIT exercise model is very clear, stating that the affective responses resulting from high-intensity exercise are perceived negatively by our brain [72-74], discouraging adherence to exercise and creating an environment of high body activation and high tension, in short, a hostile environment. However, such a scenario can be mitigated by self-selected protocols [75], but we still don't have information about this in BD patients.

With this information in hand, the HIIT exercise protocol performed by the groups of participants was prepared in such a way that we did not obtain the effects already expected in the literature. In our study, the affective responses were significantly positive for all groups, and with greater magnitude precisely due to the HIT exercise model (Continuous control: 0.2 ± 0.5 *vs.* 2.5 ± 0.5 – ES = 2.50; HIIT control: 0.7 ± 1.1 *vs.* 4.8 ± 0.3 – ES = 3.54; continuous BD: 2.0 ± 1.0 *vs.* 3.17 ± 2.2 – ES = 0.58; HIIT BD: 0.7 ± 1.2 *vs.* 4.0 ± 1.5 – ES = 2.62; respectively pre and post exercise), which goes against what is exposed in the literature [72, 73].

One of the recommendations in the literature for obtaining positive affective responses is the priority use of aerobic metabolism [70]. And unlike what is positioned by some authors, HIT fits into a transition zone between both aerobic and anaerobic systems (*crossover*) [42, 76-78]. Therefore, what was mainly adjusted in our HIT exercise protocol was to prevent the stimuli performed at high intensity from transiting into an anaerobic zone for long periods of time. This effect suggests that the fractionation of the stimuli into short periods of exercise may be an important way to maintain regular aerobic exercise and, consequently, improve health-related indicators [47, 79-81].

6.3. Secondary Outcome

We attribute some of the results obtained for anxiety to this attenuated effort response. For example, our initial hypothesis of an anxiolytic effect of exercise was accepted beyond expectations, with significant reductions observed for all groups after exercise, and a greater impact on the BD group in HIIT (3.63 ± 1.19 *vs.* 2.0 ± 1.51; and 3.38 ± 1.41 *vs.* 0.88 ± 0.99, respectively for pre and post continuous *vs.* HIIT).

It is important to emphasize that, among individuals diagnosed with BD, before the continuous exercise session, 3 participants (37.5%) were categorized as experiencing mild anxiety, while 5 (62.5%) were classified as having moderate anxiety. Following the session, this distribution shifted, with 7 individuals (87.5%) reporting mild anxiety and only 1 (12.5%) remaining in the moderate anxiety category. This adjustment was also reflected in the average scores, which moved from a pre-exercise classification of mild to moderate anxiety (3.63 ± 1.19) to a post-exercise classification of mild anxiety (2.0 ± 1.51).

As for the HIIT session, an equal number of subjects (4 each; 50%) were initially classified with mild and moderate anxiety. After the session, however, 3 individuals (37.5%) were categorized as having no anxiety, and the remaining 5 (62.5%) reported mild anxiety. Consequently, the average anxiety score shifted markedly from mild to moderate before exercise (3.38 ± 1.41) to a range between no anxiety and mild anxiety after exercise (0.88 ± 0.99).

In the control group, before the continuous session, 2 participants (25%) presented no anxiety, while 6 (75%) reported mild symptoms, an identical pattern to that observed post-session. Regarding the HIIT session, at baseline, 4 individuals (50%) were categorized as having no anxiety and 4 (50%) as having mild anxiety. Afterward, 6 participants (75%) remained in the no-anxiety category, while 2 (25%) continued to report mild symptoms. Despite these apparent shifts, the mean values were largely unaffected, with anxiety levels moving from a mild classification at baseline (1.1 ± 1.36) to a range encompassing no anxiety to mild anxiety post-exercise (0.4 ± 0.74).

This pattern can be partially interpreted through the lens of the Law of Initial Value [82], which posits that the direction and magnitude of a physiological or emotional response to a stimulus are largely dependent on the pre-stimulus level of that function. For instance, individuals beginning with lower anxiety levels are less likely to exhibit substantial reductions, whereas those with elevated baseline anxiety may show more pronounced decreases [82]. Moreover, due to the nature of the Likert scale (ranging from 0 to 10), score fluctuations occur in fixed one-point intervals, meaning that a single participant's large shift can disproportionately increase variability. This is illustrated by the standard deviation values during the HIIT session, which exceeded the mean, suggesting greater data dispersion and limiting the statistical detection of group differences. Notably, although the reduction in anxiety scores for the control group during the HIIT session did not reach statistical significance, the p-value (p = 0.063) indicates a marginal trend toward significance.

On the other hand, given that no reduction value suggests a minimal important clinical difference for the SUD anxiety scale, we suggest that future studies investigate a minimal important clinical difference value and that perhaps our findings may indicate a path for future research to establish clinical significance criteria.

Regarding studies that have investigated the effects of HIIT and its anxiolytic responses specifically in patients with BD, the literature still seems to lack information. However, when we analyzed such responses in interval stimuli with different mental disorders [38, 83], Wu *et al*. [84] demonstrated the effectiveness of a HIIT program in 22 patients after 8 weeks of exercise. They observed a significant reduction in anxiety scores "BAI - Beck Anxiety Inventory" (13.67 ± 13.83 to 10.06 ± 11.18, p = 0.003), and depression scores "BDI - Beck Depression Inventory" (19.56 ± 15.28 to 15.89 ± 14.33, p < 0.001) after training. Such outcomes are similar to other pharmacological strategies [85].

When looking at continuous moderate exercise models, different reviews and meta-analyses have already shown us relative effectiveness and a significant reduction in anxiety levels. [38, 86]. In addition, there appears to be a clear inverse association between high initial levels of anxiety and levels of physical activity, [87, 88] and, although we did not attempt to answer this question, in our study very low VO_2Max levels were found in patients with BD (control: 32.1 ± 4.9; bipolar: 25.8 ± 8.1 $mL \cdot kg^{-1} min^{-1}$), at the same time that high baseline levels of anxiety were also observed, in line with what was observed in the literature [87, 88].

Regarding the circumplex graphic models, we clearly observe that all subjects moved under the quadrant of tranquility and energy for all types of training, without a high level of stress, which configures the maintenance of positive activated affect, which suggests that both exercise models are suitable for the populations studied.

Finally, in order to try to explain how changes in autonomic nervous activity can reverberate in the responses of felt arousal, feelings scale, and anxiety, the parasympathetic nervous system, primarily mediated by the vagus nerve, plays a crucial role in maintaining emotional homeostasis. Increased parasympathetic activity is associated with enhanced heart rate variability, particularly in indices such as RMSSD, which reflect vagal tone. From a neurological perspective, enhanced vagal activity modulates the activity of key brain regions involved in mood regulation, including the prefrontal cortex, amygdala, and anterior cingulate cortex [89, 90]. These regions are directly involved in emotional regulation, executive control, and impulse moderation, functions often disrupted during manic episodes. By exerting a top-down inhibitory effect on the limbic system, increased parasympathetic tone can reduce hyperarousal and emotional reactivity, thereby contributing to a more stable mood profile and fewer affective episodes [89].

Moreover, parasympathetic activation influences neurotransmitter systems, particularly the cholinergic and serotonergic pathways, both of which are implicated in mood regulation and BD. Vagal stimulation has been shown to increase extracellular concentrations of serotonin (5-HT) in brain regions such as the dorsal raphe nucleus and prefrontal cortex, promoting antidepressant and mood-stabilizing effects [91]. Additionally, enhanced parasympathetic tone can suppress excessive dopaminergic activity in mesolimbic circuits, which is commonly associated with manic symptoms [92]. Therefore, interventions that increase vagal tone, such as aerobic exercise, may have therapeutic potential by modulating both the central autonomic network and monoaminergic transmission involved in affective disorders [93].

7. LIMITATIONS

Changes in breathing patterns could represent an intervening factor in our study, since the number of respiratory cycles per minute was not standardized, and it is known that sinus breathing would affect autonomic flow without necessarily meaning a direct response to exercise. The measured HRV was recorded in a calm and dark environment, with a controlled temperature between 21 and 23°C, with the participants' eyes closed and almost completely free from external auditory sensory stimuli. Therefore, this limitation was intentional due to the need to perform other integrated physiological analyses, which could not compete with sensory stimulation of any kind. Even in this scenario, our HRV results showed support in the scientific literature and can be considered valid [94, 95]. Furthermore, rhythmic breathing can produce strong relaxation effects, as efficient brain-heart interactions, mediated by the central autonomic network, are crucial in regulating physiological and psychological stress. The autonomic nervous system's ability to adapt to stress predicts resilience to cardiovascular, anxiety, and mood disorders. Furthermore, the present study aimed to investigate anxiety levels, as well as activation and feeling. We considered that adopting the natural breathing pattern could have a lesser effect on these variables. By excluding less relaxing breathing conditions, we sought to isolate anxiety-related brain activity and eliminate nonspecific effects related to sensorial stimulation and guided breathing [96, 97]. In addition, for the reasons mentioned above, the Guidelines for Reporting Papers on Psychiatry and Heart Rate Variability (GRAPH): recommendations for advancing research communication, encourage the application of spontaneous breathing [67].

Additionally, although our study may have limited statistical power to detect 'group × modality' interactions, and a potential for type II errors cannot be ruled out, the overall findings remain informative and contribute to understanding the observed effects. Furthermore, the study focused solely on the acute effects of exercise, without addressing long-term adaptations and the sustainability of these responses. Another limitation concerns the potential impact of confounding factors, such as the use of medications by bipolar patients, which may affect both the physiological and affective responses observed. However, the influence of psychotropic medication on acute exercise responses remains inconclusive, particularly in stable patients. Importantly, our results suggest that mood and autonomic improvements following exercise can occur even in medicated individuals, indicating that pharmacological treatment does not preclude the detection of exercise-induced benefits [98]. However, given the limited sample size, we were unable to analyze patients who were taking benzodiazepines. Therefore, these findings should be interpreted with caution, as benzodiazepines have been reported to significantly influence cardiac autonomic regulation, typically leading to a reduction in HRV through the attenuation of vagally mediated components such as HF power and RMSSD, which reflect decreased parasympathetic modulation [99]. Moreover, despite their well-established anxiolytic and sedative effects, chronic or excessive benzodiazepine use has been associated with emotional blunting and diminished affective responsiveness, suggesting possible alterations in psychoaffective regulation [100]. Collectively, these observations highlight a complex interaction between pharmacological anxiolysis and autonomic balance, wherein short-term reductions in anxiety may coexist with diminished physiological flexibility.

Finally, the highly controlled laboratory environment, characterized by silence, low luminosity, and regulated temperature, may not fully reflect the real-world conditions of exercise practice, thereby potentially compromising the ecological validity of the results.

CONCLUSION

Both continuous moderate-intensity aerobic exercise and HIIT elicited favorable acute autonomic and psychoaffective responses in BD patients and control participants. Specifically, both exercise protocols significantly led to a positive effect and reduced anxiety levels, with the HIIT modality showing a trend toward greater affective and anxiolytic benefits, particularly in the BD group. These findings support the feasibility and potential clinical utility of incorporating HIIT into exercise programs for individuals with BD, as it appears to provide a time-efficient strategy for improving autonomic function and psychoaffective states without compromising overall autonomic control.

AUTHORS' CONTRIBUTIONS

The authors confirm their contribution to the paper as follows: study conception and design: ASSF, MMS, SM; data collection: ASNA, PAI; Data Analysis or Interpretation: VA; investigation: IOS, ABP; draft manuscript: GRC, RPV, JOF, RABLM. All authors reviewed the results and approved the final version of the manuscript.

LIST OF ABBREVIATIONS

ACSM	=	American College of Sports Medicine
BD	=	Bipolar Disorder
BP	=	Blood Pressure
CI95%	=	95% Confidence Interval
CONSORT	=	Consolidated Standards of Reporting Trials
DSM-IV	=	Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition
ES	=	Effect Size
FAS	=	Felt Arousal Scale
FS	=	Feeling Scale
GRAPH	=	Guidelines for Reporting Articles on Psychiatry and Heart Rate Variability
HF	=	High Frequency
HIIT	=	High-Intensity Interval Training
HR	=	Heart Rate
HRMax	=	Maximal Heart Rate
HRRest	=	Resting Heart Rate
HRV	=	Heart Rate Variability
HRWorkload	=	Heart Rate at Workload
ICMJE	=	International Committee of Medical Journal Editors
LF	=	Low Frequency
LF/HF	=	Low-to-High Frequency Ratio
METs	=	Metabolic Equivalent (1 MET ≈ 3,5 $mL \cdot kg^{-1} \cdot min^{-1}$ de O_2)
RHR	=	Heart Rate Reserve
RMSSD	=	Root Mean Square of Successive Differences
RPE	=	Rating of Perceived Exertion (Borg 0-10)
SBP	=	Systolic Blood Pressure
SD	=	Standard Deviation
SDNN	=	Standard Deviation of NN Intervals
SUDs	=	Subjective Units of Distress Scale
TRIMP	=	Training Impulse
VLF	=	Very Low Frequency
VO_2Max	=	Maximal Oxygen Consumption

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

The study was approved by the institutional research ethics committee no. (Universidade Federal do Rio de Janeiro - #045.2015).

HUMAN AND ANIMAL RIGHTS

The human study was performed following Resolution 466/2012 of the National Health Council. All procedures were performed in accordance with the Declaration of Helsinki.

CONSENT FOR PUBLICATION

Informed consent was obtained from all subjects involved in this study.

STANDARD OF REPORTING

STROBE guidelines were followed.

AVAILABILITY OF DATA AND MATERIALS

All raw data will be available upon a reasonable request to the corresponding author.

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None.

CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

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